

# El Niño and the Ski Season

What the Oceanic Niño Index actually predicts about powder days, storm frequency and base depth across 43 ski regions of the western United States — and, just as usefully, where it predicts nothing at all.

|                                           |                                                |                                             |
|-------------------------------------------|------------------------------------------------|---------------------------------------------|
| <div>932</div> <div>SNOTEL STATIONS</div> | <div>40.2M</div> <div>DAILY OBSERVATIONS</div> | <div>1964–2026</div> <div>WATER YEARS</div> |
| <div>43</div> <div>SKI REGIONS</div>      | <div>944</div> <div>TESTS RUN</div>            | <div>46</div> <div>SURVIVE FDR</div>        |

Prepared 21 September 2026 · Data: NOAA CPC · NRCS AWDB · NCEI nClimDiv · Method after Brown & Harper (2026), HESS 30, 5735

## THE SHORT VERSION

### Four findings

1. The effect is real but geographically narrow. Of 944 tests across every region, window of winter and measure, 46 survive false-discovery control. Nearly all of them sit in the northern Rockies and the Cascades, where El Niño costs roughly a foot of snow, or in New Mexico and Arizona, where it adds snow in spring.

2. For most of the West, El Niño predicts nothing. The Wasatch, the I-70 corridor, Tahoe, the San Juans, Sun Valley and the Elk Mountains never clear significance on any measure in any window. That is a finding, not a gap.

3. The Christmas–New Year window behaves differently from the rest of winter. Every other window follows a north–south seesaw. The holidays lean toward less snow in 98 of 120 tests, in all three latitude bands at once, including the southern ranges that otherwise gain.

4. The strength of an event carries no information. Tested directly across 2508 region-window-measure combinations, nothing survives. A very strong El Niño delivers more powder days than a strong one.

RESULT

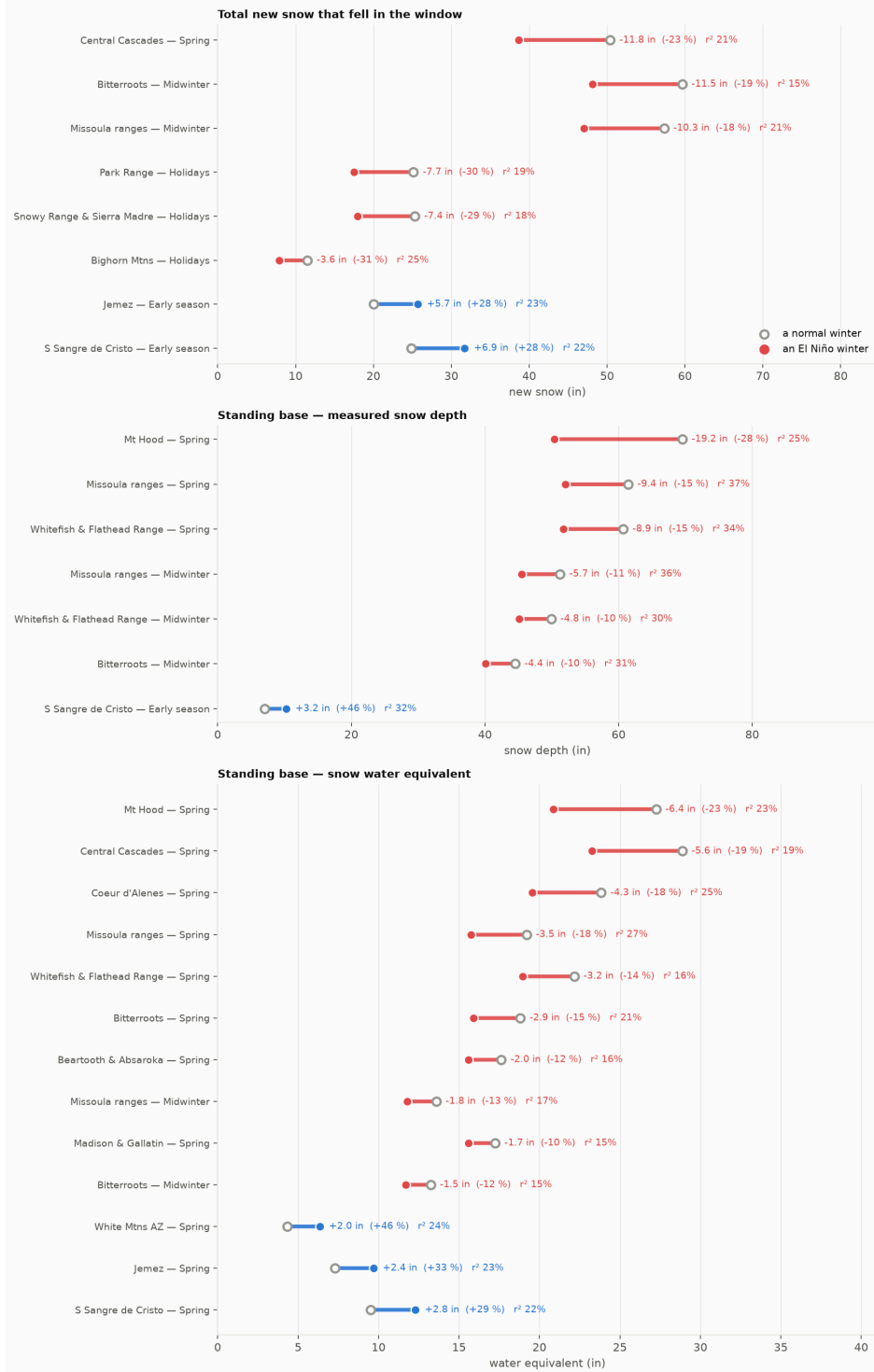
What survives, in inches

These are the region–window–measure combinations that clear Benjamini–Hochberg false-discovery control across all 944 tests. Each row compares the long-run average of every winter on record against the average of El Niño winters only.

| Inches. New snow is what fell during the window; base is what was standing, measured both as depth and as water equivalent. |              |                        |         |        |         |            |        |     |
|-----------------------------------------------------------------------------------------------------------------------------|--------------|------------------------|---------|--------|---------|------------|--------|-----|
| SKI REGION                                                                                                                  | WINDOW       | MEASURE                | WINTERS | NORMAL | EL NIÑO | DIFFERENCE | CHANGE | R²  |
| Bighorn Mtns (Meadowlark)                                                                                                   | Holidays     | Total new snow         | 48      | 11.5   | 7.9     | -3.6 in    | -31%   | 25% |
| Park Range (Steamboat)                                                                                                      | Holidays     | Total new snow         | 47      | 25.1   | 17.5    | -7.7 in    | -30%   | 19% |
| Snowy Range & Sierra Madre                                                                                                  | Holidays     | Total new snow         | 48      | 25.3   | 17.9    | -7.4 in    | -29%   | 18% |
| Mt Hood                                                                                                                     | Spring       | Base, snow depth       | 29      | 69.5   | 50.4    | -19.2 in   | -28%   | 25% |
| Mt Hood                                                                                                                     | Spring       | Base, water equivalent | 48      | 27.3   | 20.8    | -6.4 in    | -23%   | 23% |
| Central Cascades (Stevens/Snoqualmie)                                                                                       | Spring       | Total new snow         | 45      | 50.4   | 38.6    | -11.8 in   | -23%   | 21% |
| Central Cascades (Stevens/Snoqualmie)                                                                                       | Spring       | Base, water equivalent | 45      | 28.9   | 23.3    | -5.6 in    | -19%   | 19% |
| Bitterroots (Lost Trail)                                                                                                    | Midwinter    | Total new snow         | 59      | 59.7   | 48.1    | -11.5 in   | -19%   | 15% |
| Missoula ranges (Montana Snowbowl)                                                                                          | Spring       | Base, water equivalent | 46      | 19.2   | 15.7    | -3.5 in    | -18%   | 27% |
| Coeur d'Alenes (Silver/Lookout)                                                                                             | Spring       | Base, water equivalent | 44      | 23.8   | 19.6    | -4.3 in    | -18%   | 25% |
| Missoula ranges (Montana Snowbowl)                                                                                          | Midwinter    | Total new snow         | 46      | 57.3   | 47.1    | -10.3 in   | -18%   | 21% |
| Bitterroots (Lost Trail)                                                                                                    | Spring       | Base, water equivalent | 59      | 18.8   | 15.9    | -2.9 in    | -15%   | 21% |
| Missoula ranges (Montana Snowbowl)                                                                                          | Spring       | Base, snow depth       | 25      | 61.4   | 52.0    | -9.4 in    | -15%   | 37% |
| Whitefish & Flathead Range                                                                                                  | Spring       | Base, snow depth       | 25      | 60.7   | 51.8    | -8.9 in    | -15%   | 34% |
| Whitefish & Flathead Range                                                                                                  | Spring       | Base, water equivalent | 52      | 22.2   | 19.0    | -3.2 in    | -14%   | 16% |
| Missoula ranges (Montana Snowbowl)                                                                                          | Midwinter    | Base, water equivalent | 46      | 13.6   | 11.8    | -1.8 in    | -13%   | 17% |
| Bitterroots (Lost Trail)                                                                                                    | Midwinter    | Base, water equivalent | 59      | 13.2   | 11.7    | -1.5 in    | -12%   | 15% |
| Beartooth & Absaroka (Red Lodge)                                                                                            | Spring       | Base, water equivalent | 53      | 17.6   | 15.6    | -2.0 in    | -12%   | 16% |
| Missoula ranges (Montana Snowbowl)                                                                                          | Midwinter    | Base, snow depth       | 25      | 51.2   | 45.5    | -5.7 in    | -11%   | 36% |
| Bitterroots (Lost Trail)                                                                                                    | Midwinter    | Base, snow depth       | 25      | 44.5   | 40.1    | -4.4 in    | -10%   | 31% |
| Madison & Gallatin (Big Sky)                                                                                                | Spring       | Base, water equivalent | 60      | 17.2   | 15.6    | -1.7 in    | -10%   | 15% |
| Whitefish & Flathead Range                                                                                                  | Midwinter    | Base, snow depth       | 25      | 49.9   | 45.1    | -4.8 in    | -10%   | 30% |
| S Sangre de Cristo (Santa Fe)                                                                                               | Early season | Total new snow         | 46      | 24.8   | 31.7    | +6.9 in    | +28%   | 22% |
| Jemez (Pajarito)                                                                                                            | Early season | Total new snow         | 46      | 20.0   | 25.7    | +5.7 in    | +28%   | 23% |
| S Sangre de Cristo (Santa Fe)                                                                                               | Spring       | Base, water equivalent | 46      | 9.5    | 12.3    | +2.8 in    | +29%   | 22% |

| SKI REGION                    | WINDOW       | MEASURE                | WINTERS | NORMAL | EL NIÑO | DIFFERENCE | CHANGE | R²  |
|-------------------------------|--------------|------------------------|---------|--------|---------|------------|--------|-----|
| Jemez (Pajarito)              | Spring       | Base, water equivalent | 46      | 7.3    | 9.7     | +2.4 in    | +33%   | 23% |
| White Mtns AZ (Sunrise Park)  | Spring       | Base, water equivalent | 46      | 4.3    | 6.3     | +2.0 in    | +46%   | 24% |
| S Sangre de Cristo (Santa Fe) | Early season | Base, snow depth       | 25      | 7.0    | 10.2    | +3.2 in    | +46%   | 32% |

### What El Niño changes, in inches — only results that survive false-discovery control



Only the results that survive false-discovery control, in inches. Three separate axes: inches of falling snow, inches of standing snow depth and inches of water in that snow are different quantities and cannot share a scale. The open circle is a normal winter; the filled circle is an El Niño winter.

## RESULT

# What survives, in days

| Days within the window. A powder day is a day with at least six inches of new snow; a storm day at least two. |              |                          |         |        |         |            |        |     |
|---------------------------------------------------------------------------------------------------------------|--------------|--------------------------|---------|--------|---------|------------|--------|-----|
| SKI REGION                                                                                                    | WINDOW       | MEASURE                  | WINTERS | NORMAL | EL NIÑO | DIFFERENCE | CHANGE | R²  |
| Missoula ranges (Montana Snowbowl)                                                                            | Midwinter    | Powder days (≥6 in)      | 46      | 1.4    | 0.8     | -0.6 d     | -42%   | 31% |
| Bighorn Mtns (Meadowlark)                                                                                     | Holidays     | Storm days (≥2 in)       | 48      | 1.7    | 1.0     | -0.7 d     | -40%   | 20% |
| Park Range (Steamboat)                                                                                        | Holidays     | Storm days (≥2 in)       | 47      | 4.3    | 2.7     | -1.6 d     | -36%   | 22% |
| Coeur d'Alenes (Silver/Lookout)                                                                               | Midwinter    | Powder days (≥6 in)      | 44      | 2.0    | 1.3     | -0.7 d     | -34%   | 22% |
| Snowy Range & Sierra Madre                                                                                    | Holidays     | Storm days (≥2 in)       | 48      | 4.2    | 2.9     | -1.3 d     | -32%   | 17% |
| Central Cascades (Stevens/Snoqualmie)                                                                         | Spring       | Storm days (≥2 in)       | 45      | 8.5    | 6.4     | -2.1 d     | -25%   | 24% |
| Mt Hood                                                                                                       | Spring       | Storm days (≥2 in)       | 48      | 8.5    | 7.0     | -1.5 d     | -18%   | 15% |
| Madison & Gallatin (Big Sky)                                                                                  | Midwinter    | Days with a skiable base | 60      | 32.0   | 26.9    | -5.1 d     | -16%   | 13% |
| Missoula ranges (Montana Snowbowl)                                                                            | Midwinter    | Days with a skiable base | 46      | 28.9   | 24.9    | -4.0 d     | -14%   | 25% |
| Whitefish & Flathead Range                                                                                    | Spring       | Days with a skiable base | 52      | 38.1   | 33.2    | -4.9 d     | -13%   | 18% |
| Missoula ranges (Montana Snowbowl)                                                                            | Spring       | Days with a skiable base | 46      | 32.6   | 28.5    | -4.1 d     | -13%   | 33% |
| Wind River Range                                                                                              | Spring       | Powder days (≥6 in)      | 48      | 0.6    | 0.8     | +0.1 d     | +21%   | 19% |
| White Mtns AZ (Sunrise Park)                                                                                  | Spring       | Storm days (≥2 in)       | 46      | 2.3    | 2.9     | +0.6 d     | +24%   | 17% |
| S Sangre de Cristo (Santa Fe)                                                                                 | Early season | Storm days (≥2 in)       | 46      | 3.9    | 4.9     | +1.0 d     | +26%   | 19% |
| Jemez (Pajarito)                                                                                              | Early season | Storm days (≥2 in)       | 46      | 3.4    | 4.3     | +0.9 d     | +28%   | 20% |
| S Sangre de Cristo (Santa Fe)                                                                                 | Spring       | Days with a skiable base | 46      | 22.1   | 33.3    | +11.2 d    | +51%   | 29% |
| Jemez (Pajarito)                                                                                              | Spring       | Days with a skiable base | 46      | 15.2   | 23.9    | +8.7 d     | +57%   | 21% |
| White Mtns AZ (Sunrise Park)                                                                                  | Spring       | Days with a skiable base | 46      | 8.2    | 14.3    | +6.1 d     | +74%   | 18% |

### What El Niño changes, in days — only results that survive false-discovery control



The same surviving results counted in days rather than inches.

## READING THE LARGEST EFFECTS

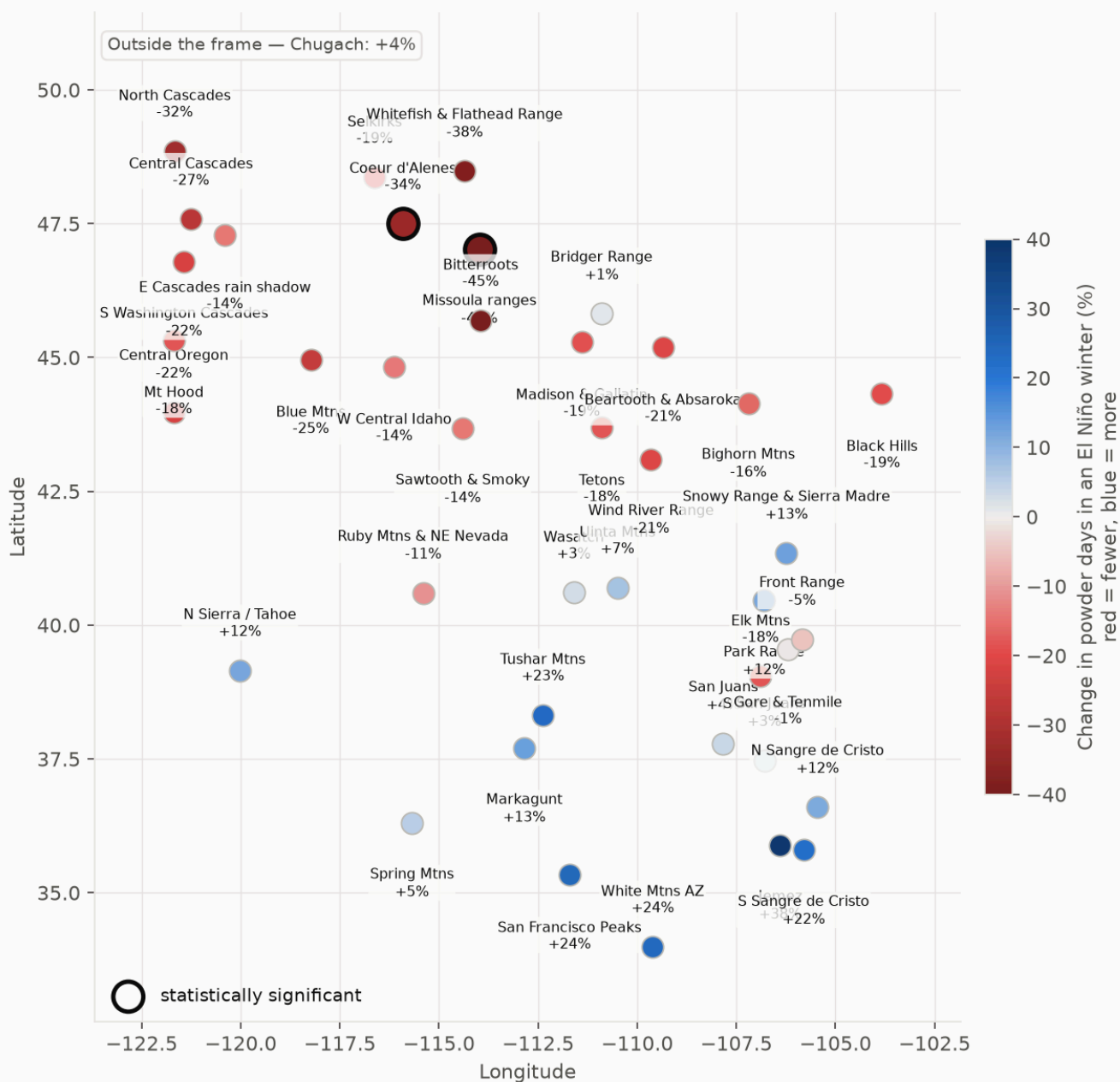
Montana Snowbowl in midwinter is the single most affected place in the study: twelve inches less new snow, two fewer storm days, and 45 % fewer powder days. The largest gain anywhere is northern New Mexico in spring, which picks up nearly nine additional days with a skiable base.

## GEOGRAPHY

### Where the effect lives

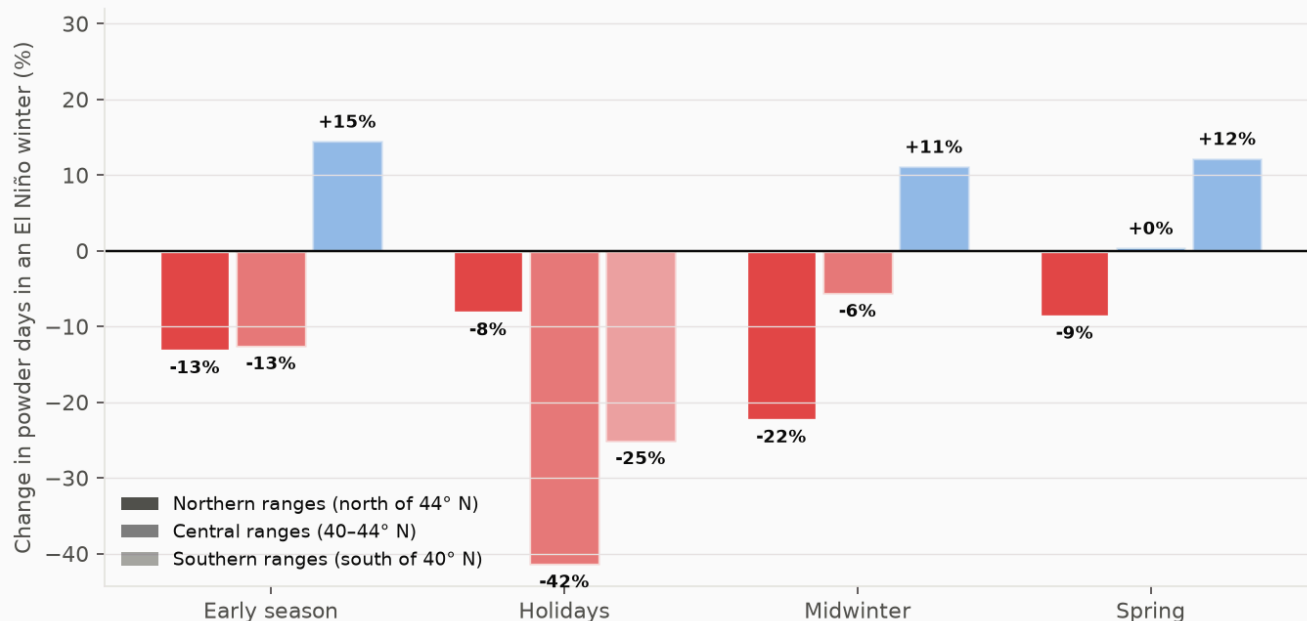
The turnover sits near 40–42° N, running through central Colorado and Utah. North of it El Niño suppresses snowfall; south of it, in spring, it adds. The southern half of the seesaw is weaker and less reliable than the northern half — several southern regions show a positive tendency that never reaches significance.

## Where El Niño costs powder days — midwinter



Change in midwinter powder days by region. Ringed markers survive false-discovery control.

El Niño's effect on powder days, by window of winter and latitude

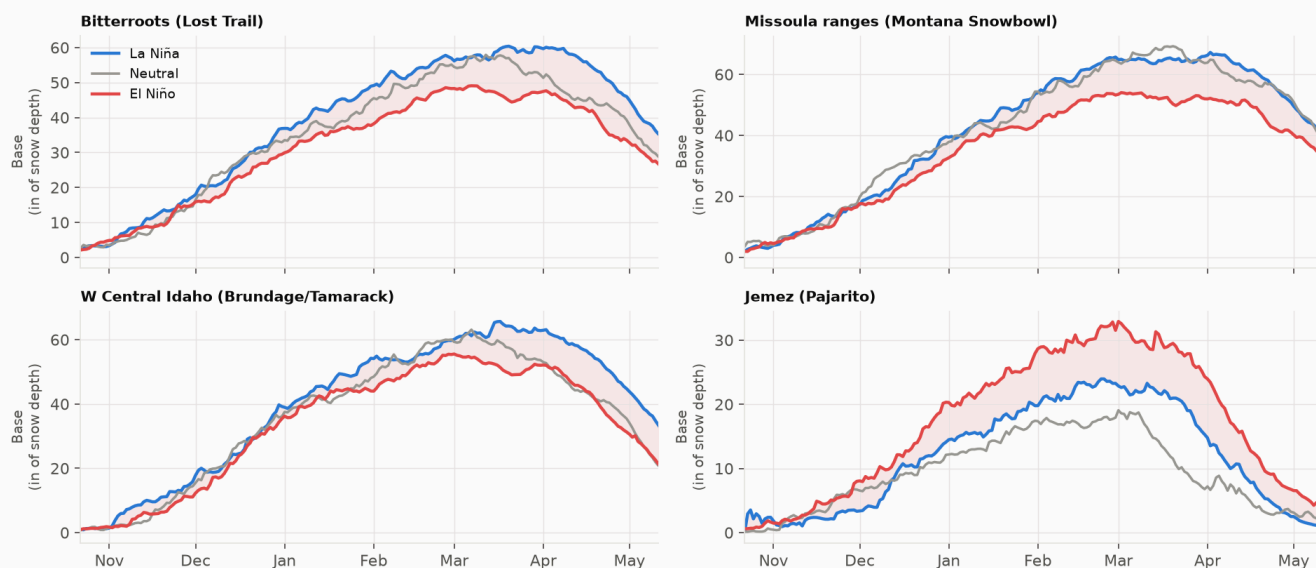


Change in powder days by window of winter, split at the ENSO node. Averaging all regions together would cancel the seesaw to near zero and say nothing true about anywhere; splitting makes both halves visible — and exposes the holiday window, which is negative in all three bands at once.

Direction consistency by window. Each test is one region and one measure; the sign test asks whether the direction holds across independent regions, which is the question that matters when each individual correlation is modest.

| WINDOW OF WINTER | TESTS | LEANING TO LESS SNOW | SHARE | MEAN R | SIGN-TEST P |
|------------------|-------|----------------------|-------|--------|-------------|
| Early season     | 120   | 79 of 120            | 66%   | -0.037 | 6.7e-04     |
| Holidays         | 120   | 98 of 120            | 82%   | -0.126 | 1.2e-12     |
| Midwinter        | 120   | 90 of 120            | 75%   | -0.108 | 3.8e-08     |
| Spring           | 120   | 70 of 120            | 58%   | -0.051 | 8.2e-02     |

### How the base builds through the winter, by ENSO phase worst-hit regions at the top, a region that gains at the bottom right



Base depth through the winter by ENSO phase, in inches of measured snow depth. Two worst-hit regions at the top, a near-neutral region and a gaining region below.

## GEOGRAPHY WITH UNCERTAINTY

### The effect moves through the winter

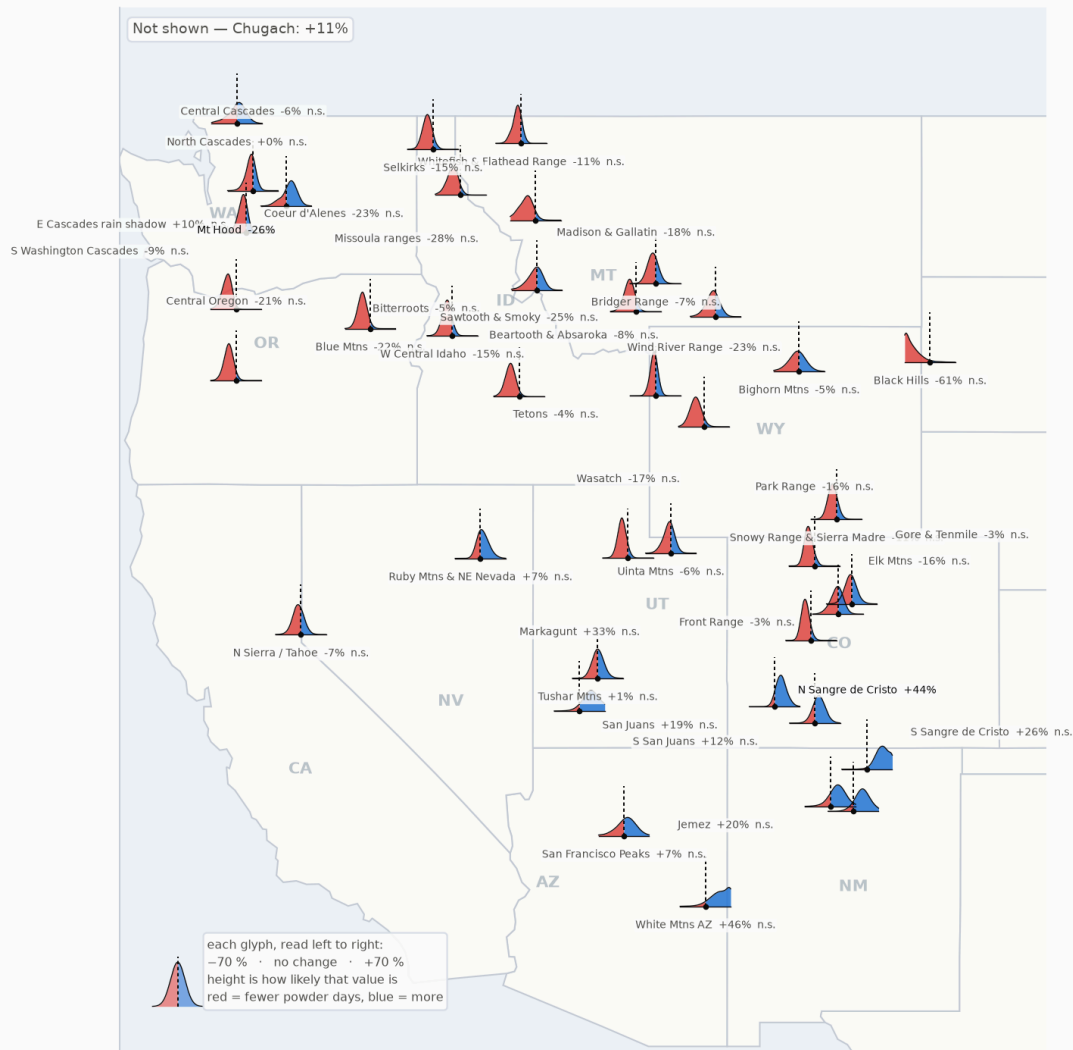
A single number per region hides how firmly it is pinned, so every ski region here carries its whole probability distribution rather than a colour. The dashed hairline through each glyph is no change. A density astride that line is a region where El Niño does nothing measurable.

Drawn window by window, the maps show what a season-long average cannot: **the effect does not sit still**. It is nearly absent in the early season, seizes the interior ranges over the holidays, moves to the northern Rockies and Cascades in midwinter, and retreats to the Cascades alone by spring. A region can be firmly hit in one window and indistinguishable from zero in the next.

#### Early season · 1 November – 15 December

Almost nothing is decided. Whatever El Niño does to a season has barely begun by mid-December.

## The change in powder days in every ski region, with its uncertainty — early season



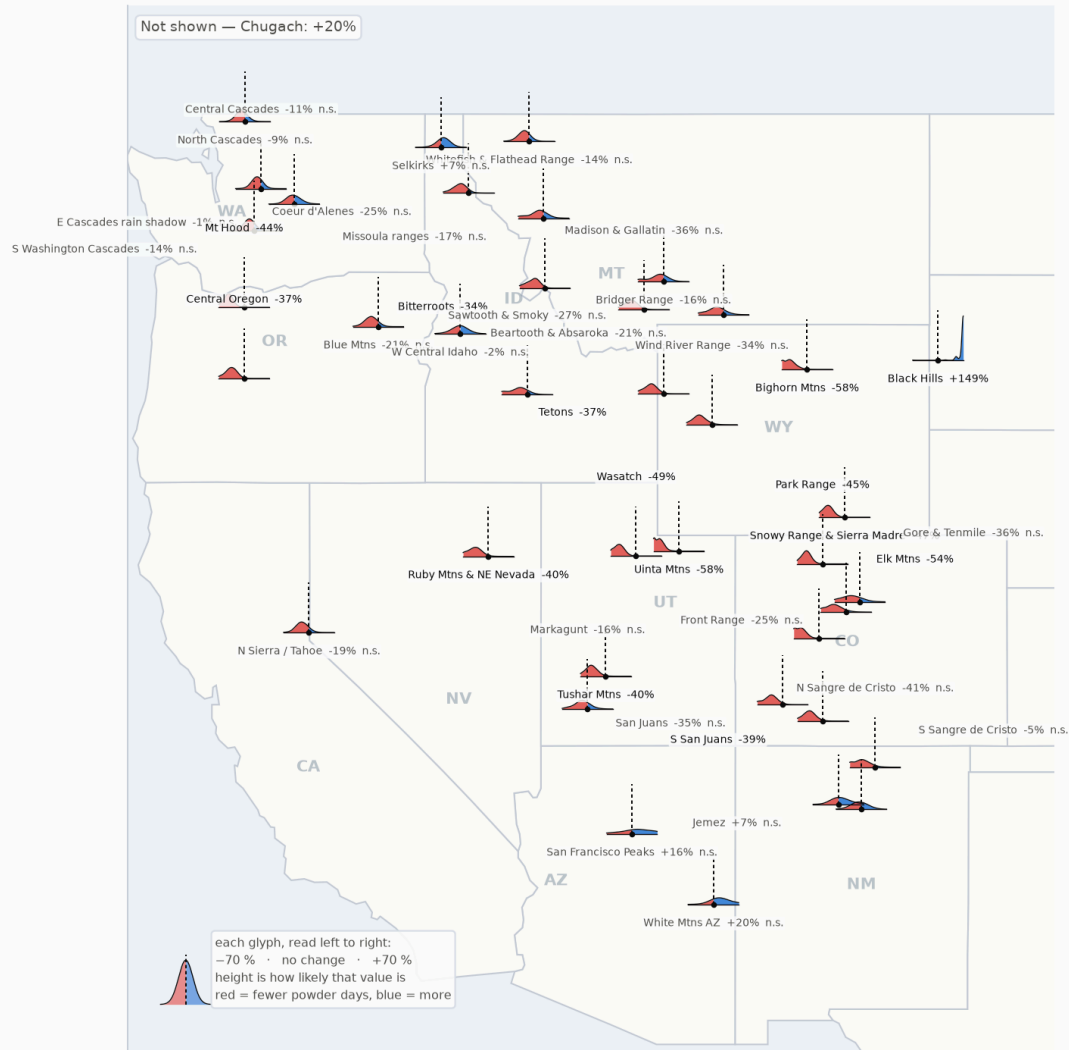
Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 37 of 39 regions straddle zero and are marked n.s.

2 of 40 regions clear zero.

## Holidays · 16 December – 5 January

The widest, most coherent signal of the winter, and it lands on the interior: the Wasatch, the Uintas, the Elk Mountains, the Park and Snowy ranges, the Rubies. These are the same regions that show nothing in midwinter.

### The change in powder days in every ski region, with its uncertainty — holidays



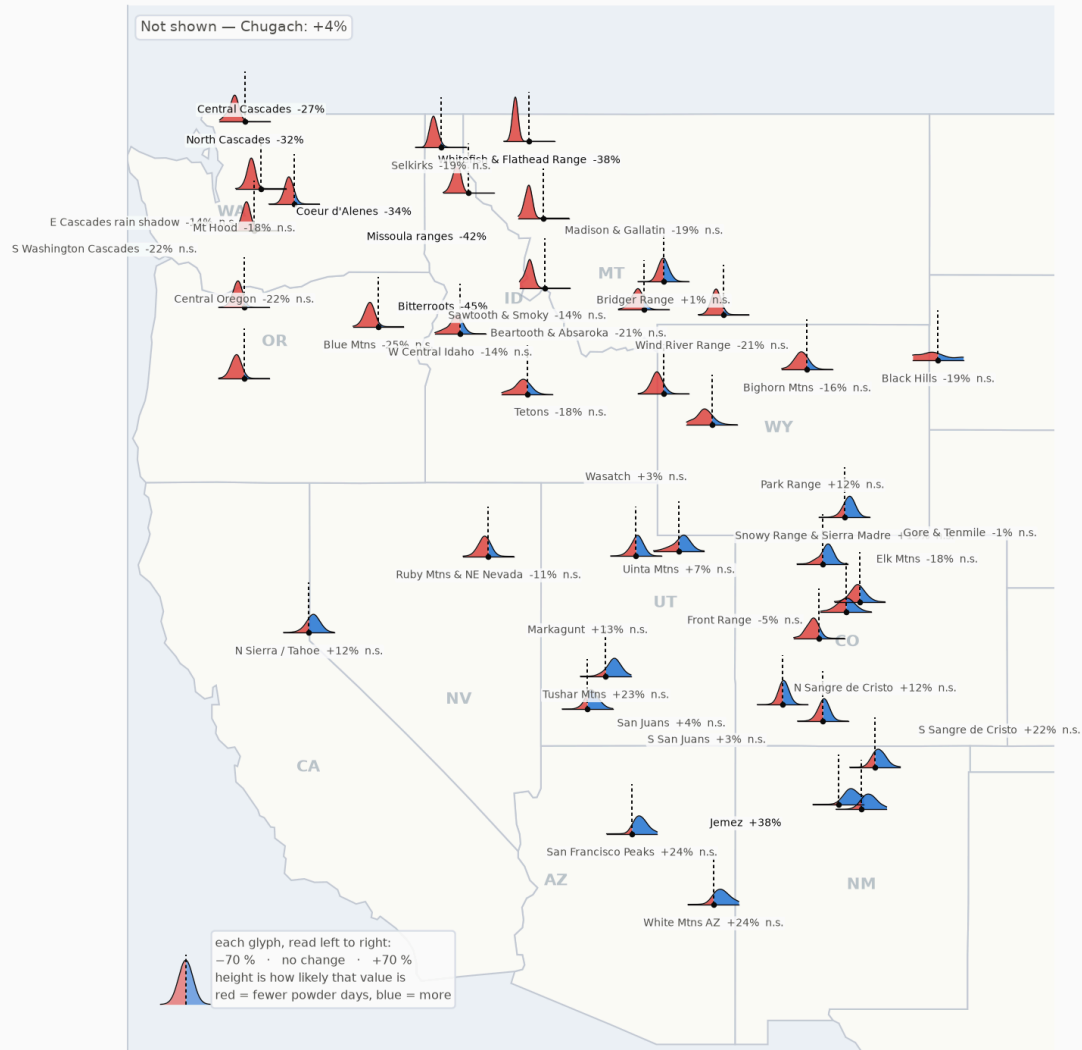
Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 25 of 39 regions straddle zero and are marked n.s.

14 of 40 regions clear zero, and the cluster sits over the Great Basin and the Colorado and Wyoming ranges rather than the north.

### Midwinter · 6 January – 28 February

The signal migrates north, to the Bitterroots, the Missoula ranges, the Coeur d'Alenes, Whitefish and the Washington Cascades, with northern New Mexico gaining. The interior ranges that were firmly hit over the holidays are now undecided.

## The change in powder days in every ski region, with its uncertainty — midwinter



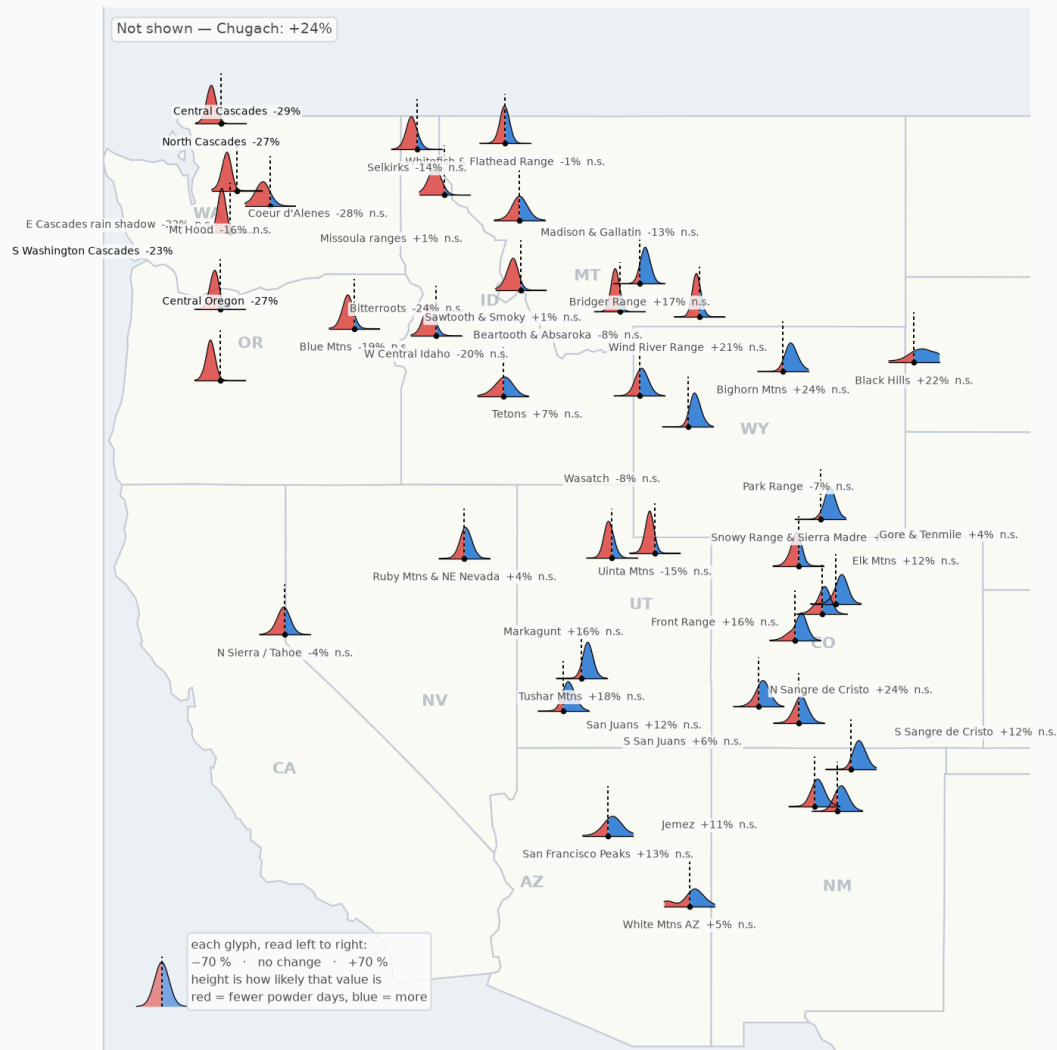
Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 32 of 39 regions straddle zero and are marked n.s.

7 of 40 regions clear zero.

**Spring · 1 March – 15 April**

Only the Cascades and central Oregon remain decided.

## The change in powder days in every ski region, with its uncertainty — spring



Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 35 of 39 regions straddle zero and are marked n.s.

4 of 40 regions clear zero, all in the Cascades and Oregon.

## TWO STANDARDS OF EVIDENCE, AND WHY THEY DISAGREE

These maps judge each region on its own: does its 95 % interval exclude zero? The headline tables apply Benjamini–Hochberg correction across all 944 tests at once, which is far stricter, because asking a question of 40 regions in four windows on six measures will turn up apparent effects by chance alone.

That is why a region can appear here at -50 % over the holidays and not appear in the list of results that survive correction. The per-region interval answers "what does the record say about *this* range"; the corrected grid answers "which of these findings would I bet on". A region that clears the first but not the second is worth watching, not worth relying on.

Every region whose 95 % interval excludes zero, by window. Percentages of powder days sit next to the inches of new snow behind them, because a twenty-one-day window has few powder days anywhere and a large percentage can rest on a

| fraction of a day. |                                       |                        |        |                     |            |            |                  |  |
|--------------------|---------------------------------------|------------------------|--------|---------------------|------------|------------|------------------|--|
| WINDOW             | SKI REGION                            | POWDER DAYS,<br>NORMAL | CHANGE | NEW SNOW,<br>NORMAL | EL<br>NIÑO | DIFFERENCE | 95 %<br>INTERVAL |  |
| Early season       | Mt Hood                               | 2.30                   | -26%   | 51 in               | 42 in      | -10 in     | -54% to -3%      |  |
|                    | N Sangre de Cristo                    | 0.62                   | +44%   | 26 in               | 32 in      | +6 in      | +7% to +85%      |  |
| Holidays           | Bighorn Mtns <small>TINY BASE</small> | 0.10                   | -58%   | 12 in               | 8 in       | -4 in      | -120% to -14%    |  |
|                    | Uinta Mtns                            | 0.54                   | -58%   | 17 in               | 11 in      | -6 in      | -90% to -29%     |  |
|                    | Elk Mtns                              | 0.57                   | -54%   | 20 in               | 15 in      | -6 in      | -89% to -22%     |  |
|                    | Wasatch                               | 0.99                   | -49%   | 23 in               | 15 in      | -8 in      | -80% to -20%     |  |
|                    | Snowy Range & Sierra Madre            | 0.79                   | -47%   | 25 in               | 18 in      | -7 in      | -77% to -18%     |  |
|                    | Park Range                            | 0.85                   | -45%   | 25 in               | 17 in      | -8 in      | -72% to -19%     |  |
|                    | Mt Hood                               | 1.53                   | -44%   | 33 in               | 26 in      | -7 in      | -72% to -19%     |  |
|                    | Tushar Mtns                           | 0.69                   | -40%   | 18 in               | 16 in      | -2 in      | -72% to -8%      |  |
|                    | Ruby Mtns & NE Nevada                 | 0.65                   | -40%   | 19 in               | 14 in      | -5 in      | -83% to -3%      |  |
|                    | S San Juans                           | 1.03                   | -39%   | 22 in               | 18 in      | -4 in      | -74% to -4%      |  |
|                    | Tetons                                | 0.91                   | -37%   | 26 in               | 20 in      | -7 in      | -77% to -3%      |  |
|                    | Central Oregon                        | 1.56                   | -37%   | 35 in               | 28 in      | -6 in      | -71% to -7%      |  |
|                    | Bitterroots                           | 0.73                   | -34%   | 24 in               | 20 in      | -4 in      | -76% to -3%      |  |
|                    | Black Hills <small>TINY BASE</small>  | 0.11                   | +149%  | 10 in               | 11 in      | +1 in      | +50% to +355%    |  |
| Midwinter          | Bitterroots                           | 1.60                   | -45%   | 60 in               | 48 in      | -12 in     | -71% to -22%     |  |
|                    | Missoula ranges                       | 1.45                   | -42%   | 57 in               | 47 in      | -10 in     | -64% to -24%     |  |
|                    | Whitefish & Flathead Range            | 1.55                   | -38%   | 59 in               | 48 in      | -11 in     | -53% to -24%     |  |
|                    | Coeur d'Alenes                        | 2.02                   | -34%   | 63 in               | 53 in      | -10 in     | -58% to -13%     |  |
|                    | North Cascades                        | 4.63                   | -32%   | 103 in              | 85 in      | -18 in     | -63% to -8%      |  |
|                    | Central Cascades                      | 3.62                   | -27%   | 84 in               | 71 in      | -13 in     | -51% to -7%      |  |
|                    | Jemez                                 | 0.74                   | +38%   | 30 in               | 37 in      | +7 in      | +0% to +85%      |  |
| Spring             | Central Cascades                      | 1.71                   | -29%   | 50 in               | 39 in      | -12 in     | -56% to -6%      |  |
|                    | North Cascades                        | 3.19                   | -27%   | 79 in               | 67 in      | -12 in     | -53% to -2%      |  |
|                    | Central Oregon                        | 1.90                   | -27%   | 54 in               | 46 in      | -9 in      | -52% to -3%      |  |
|                    | S Washington Cascades                 | 2.08                   | -23%   | 60 in               | 50 in      | -11 in     | -46% to -3%      |  |

THE NEGATIVE RESULT

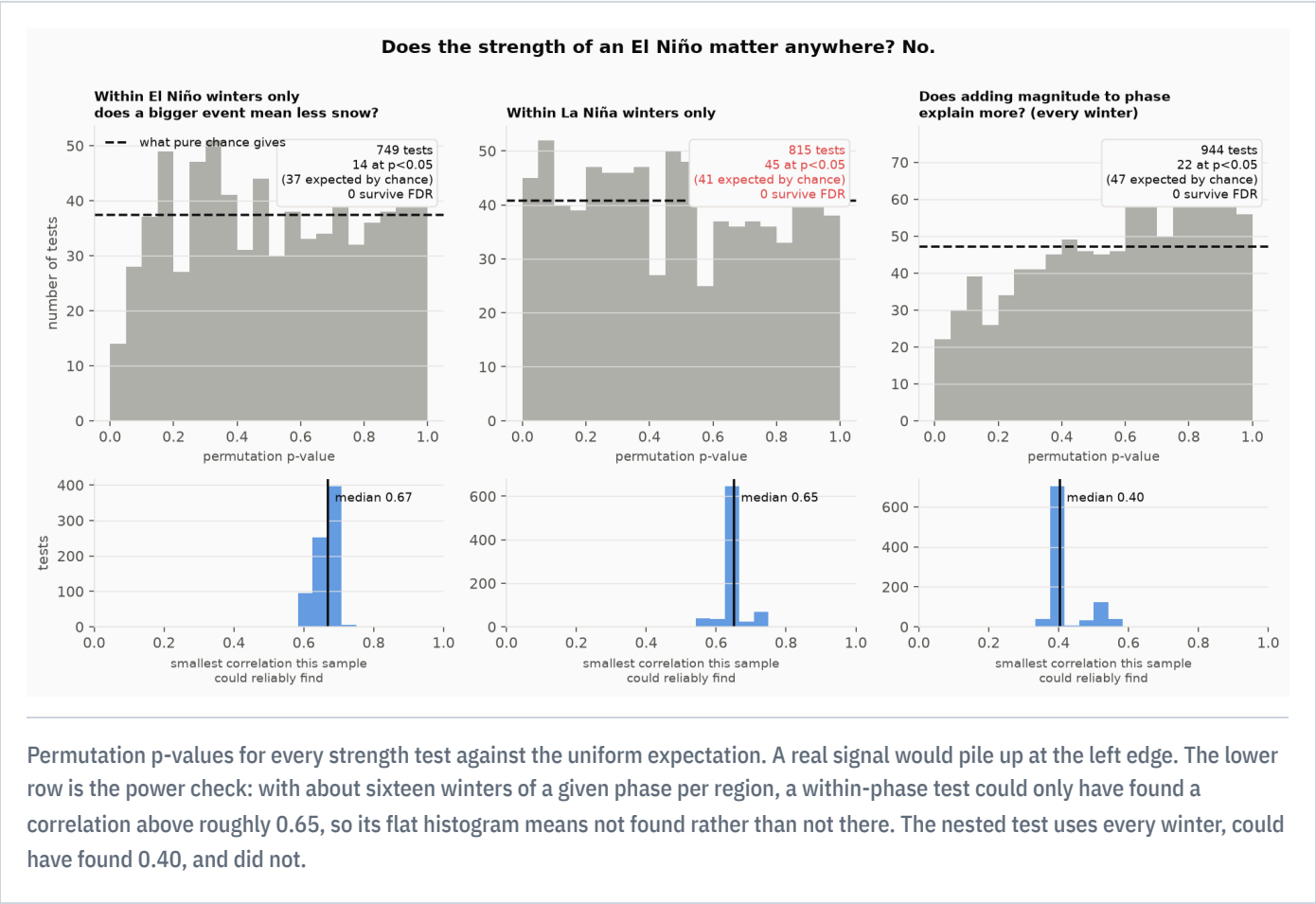
Event strength carries no information

Phase says which way a winter leans. Strength says how far the index went. The two are routinely conflated in seasonal forecasting, so strength was tested head-on rather than inferred: every region, window and measure was asked whether a larger event means less snow among El Niño winters only, the same among La Niña winters, and whether adding the index value to a phase-only model explains more.

| 2508 strength tests with permutation p-values and false-discovery control across the whole family. |           |          |                    |             |                     |  |
|----------------------------------------------------------------------------------------------------|-----------|----------|--------------------|-------------|---------------------|--|
| TEST                                                                                               | TESTS RUN | P < 0.05 | EXPECTED BY CHANCE | SURVIVE FDR | SMALLEST R FINDABLE |  |
| Within El Niño winters only                                                                        | 749       | 14       | 37                 | 0           | 0.67                |  |

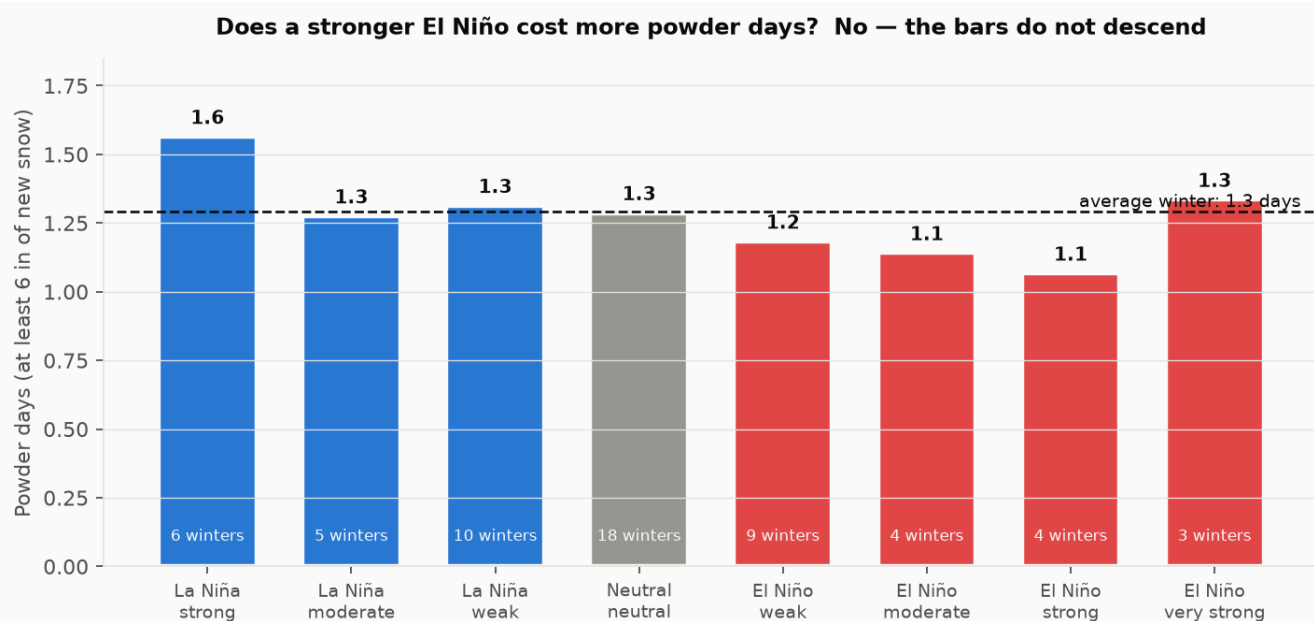
| TEST                                     | TESTS RUN | P < 0.05 | EXPECTED BY CHANCE | SURVIVE FDR | SMALLEST R FINDABLE |
|------------------------------------------|-----------|----------|--------------------|-------------|---------------------|
| Within La Niña winters only              | 815       | 45       | 41                 | 0           | 0.65                |
| Adding magnitude to phase (every winter) | 944       | 22       | 47                 | 0           | 0.40                |

Nothing survives. The p-values are distributed exactly as pure chance distributes them, and the within-El Niño family returns *fewer* nominal hits than chance alone would produce.



Midwinter powder days by strength bin, averaged over every region. If magnitude mattered the bars would descend from left to right.

| PHASE AND STRENGTH    | WINTERS | POWDER DAYS |
|-----------------------|---------|-------------|
| La Niña · strong      | 6       | 1.56        |
| La Niña · moderate    | 5       | 1.27        |
| La Niña · weak        | 10      | 1.31        |
| Neutral · neutral     | 18      | 1.29        |
| El Niño · weak        | 9       | 1.18        |
| El Niño · moderate    | 4       | 1.14        |
| El Niño · strong      | 4       | 1.07        |
| El Niño · very strong | 3       | 1.33        |



Variance decomposition. Knowing only the phase beats knowing the index value; adding magnitude makes the prediction worse in every window.

## PRACTICAL CONSEQUENCE

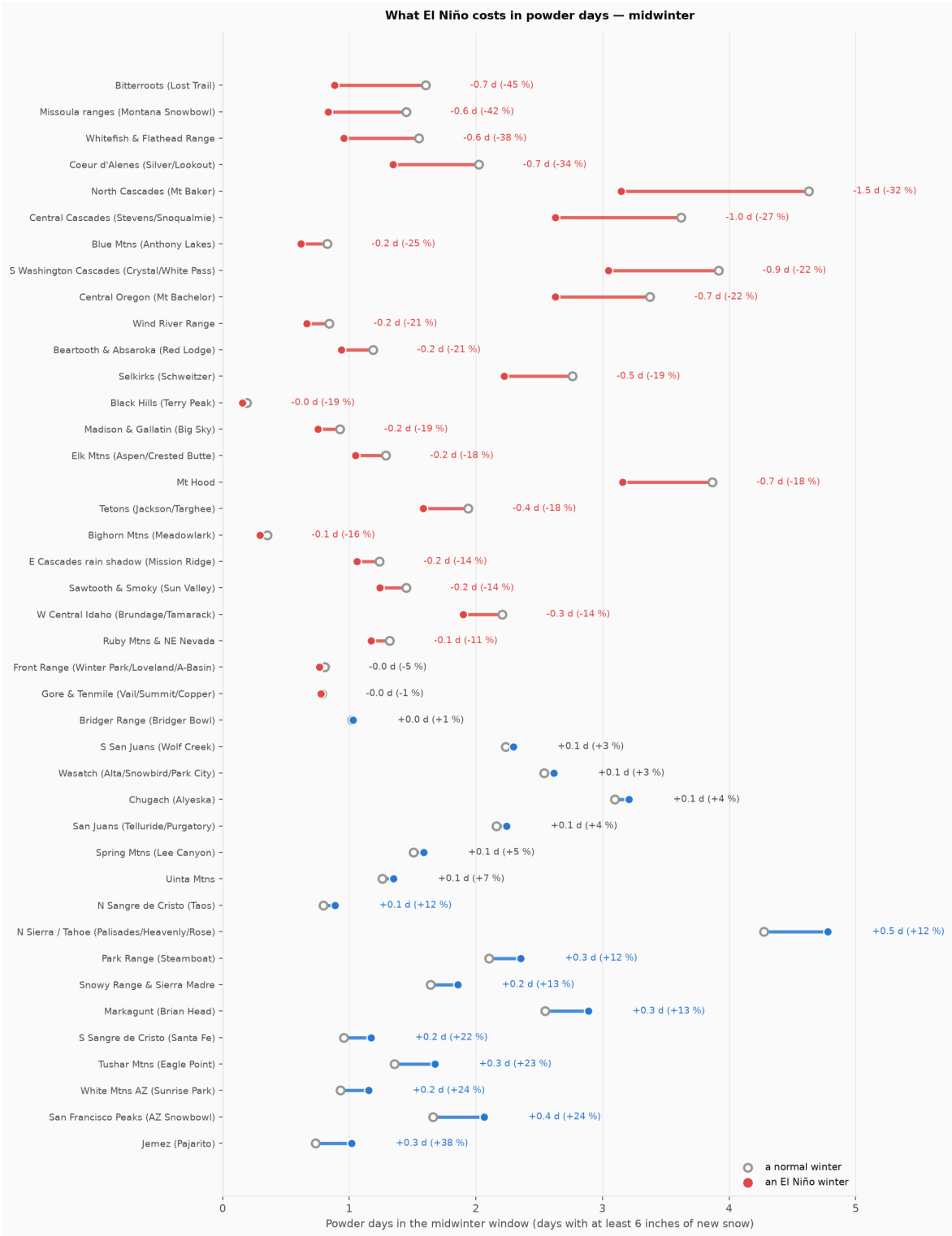
## FULL RESULTS

The complete picture, including the regions where nothing is detectable. Ordered from the largest loss to the largest gain. Regions with under one powder day a month show large percentage swings on very small absolute differences — read the day counts, not the percentages, at the extremes.

Powder days in the midwinter window, 6 January to 28 February.

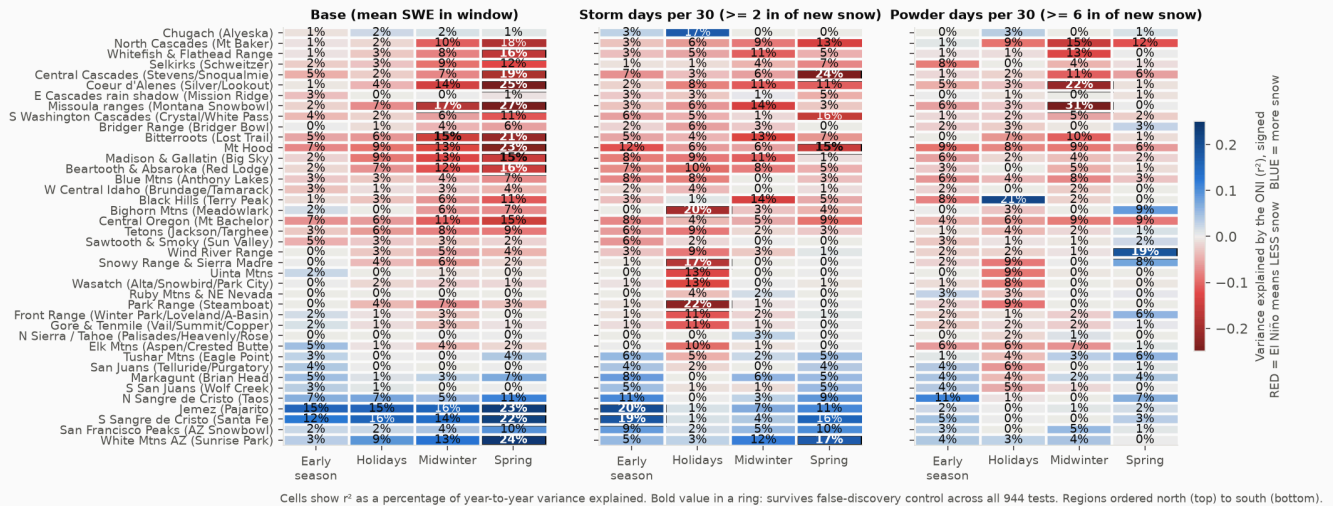
| Ski Region                                 | States | Snow Climate  | Winters | Normal | El Niño | Diff | Change | La Niña | Status                     |
|--------------------------------------------|--------|---------------|---------|--------|---------|------|--------|---------|----------------------------|
| Bitterroots (Lost Trail)                   | MT+ID  | transitional  | 59      | 1.6    | 0.9     | -0.7 | -45%   | 2.3     | <div>SURVIVES FDR</div>    |
| Missoula ranges (Montana Snowbowl)         | MT     | transitional  | 46      | 1.4    | 0.8     | -0.6 | -42%   | 2.0     | <div>SURVIVES FDR</div>    |
| Whitefish & Flathead Range                 | MT     | transitional  | 52      | 1.5    | 1.0     | -0.6 | -38%   | 2.0     | <div>SURVIVES FDR</div>    |
| Coeur d'Alenes (Silver/Lookout)            | ID+MT  | transitional  | 44      | 2.0    | 1.3     | -0.7 | -34%   | 2.7     | <div>SURVIVES FDR</div>    |
| North Cascades (Mt Baker)                  | WA     | maritime      | 31      | 4.6    | 3.1     | -1.5 | -32%   | 5.8     | <div>NOT SIGNIFICANT</div> |
| Central Cascades (Stevens/Snoqualmie)      | WA     | maritime      | 45      | 3.6    | 2.6     | -1.0 | -27%   | 4.3     | <div>SURVIVES FDR</div>    |
| Blue Mtns (Anthony Lakes)                  | OR     | transitional  | 48      | 0.8    | 0.6     | -0.2 | -25%   | 0.9     | <div>NOT SIGNIFICANT</div> |
| S Washington Cascades (Crystal/White Pass) | WA     | maritime      | 45      | 3.9    | 3.0     | -0.9 | -22%   | 4.3     | <div>NOT SIGNIFICANT</div> |
| Central Oregon (Mt Bachelor)               | OR     | transitional  | 46      | 3.4    | 2.6     | -0.7 | -22%   | 4.2     | <div>NOT SIGNIFICANT</div> |
| Wind River Range                           | WY     | continental   | 48      | 0.8    | 0.7     | -0.2 | -21%   | 0.7     | <div>SURVIVES FDR</div>    |
| Beartooth & Absaroka (Red Lodge)           | MT+WY  | continental   | 53      | 1.2    | 0.9     | -0.2 | -21%   | 1.5     | <div>SURVIVES FDR</div>    |
| Selkirks (Schweitzer)                      | ID     | transitional  | 43      | 2.8    | 2.2     | -0.5 | -19%   | 3.3     | <div>NOT SIGNIFICANT</div> |
| Black Hills (Terry Peak)                   | SD     | continental   | 26      | 0.2    | 0.2     | -0.0 | -19%   | 0.3     | <div>NOT SIGNIFICANT</div> |
| Madison & Gallatin (Big Sky)               | MT     | continental   | 60      | 0.9    | 0.8     | -0.2 | -19%   | 1.0     | <div>SURVIVES FDR</div>    |
| Elk Mtns (Aspen/Crested Butte)             | CO     | continental   | 46      | 1.3    | 1.1     | -0.2 | -18%   | 1.5     | <div>NOT SIGNIFICANT</div> |
| Mt Hood                                    | OR     | maritime      | 48      | 3.9    | 3.2     | -0.7 | -18%   | 4.7     | <div>SURVIVES FDR</div>    |
| Tetons (Jackson/Targhee)                   | WY+ID  | intermountain | 46      | 1.9    | 1.6     | -0.4 | -18%   | 1.8     | <div>NOT SIGNIFICANT</div> |
| Bighorn Mtns (Meadowlark)                  | WY     | continental   | 48      | 0.4    | 0.3     | -0.1 | -16%   | 0.4     | <div>SURVIVES FDR</div>    |
| E Cascades rain shadow (Mission Ridge)     | WA     | transitional  | 45      | 1.2    | 1.1     | -0.2 | -14%   | 1.5     | <div>NOT SIGNIFICANT</div> |
| Sawtooth & Smoky (Sun Valley)              | ID     | intermountain | 46      | 1.4    | 1.2     | -0.2 | -14%   | 1.3     | <div>NOT SIGNIFICANT</div> |
| W Central Idaho (Brundage/Tamarack)        | ID     | intermountain | 46      | 2.2    | 1.9     | -0.3 | -14%   | 2.3     | <div>NOT SIGNIFICANT</div> |
| Ruby Mtns & NE Nevada                      | NV     | continental   | 46      | 1.3    | 1.2     | -0.1 | -11%   | 1.4     | <div>NOT SIGNIFICANT</div> |
| Front Range (Winter Park/Loveland/A-Basin) | CO     | continental   | 48      | 0.8    | 0.8     | -0.0 | -5%    | 0.8     | <div>NOT SIGNIFICANT</div> |
| Gore & Tenmile (Vail/Summit/Copper)        | CO     | continental   | 48      | 0.8    | 0.8     | -0.0 | -1%    | 0.8     | <div>NOT SIGNIFICANT</div> |
| Bridger Range (Bridger Bowl)               | MT     | continental   | 50      | 1.0    | 1.0     | +0.0 | +1%    | 1.0     | <div>NOT SIGNIFICANT</div> |
| S San Juans (Wolf Creek)                   | CO     | continental   | 46      | 2.2    | 2.3     | +0.1 | +3%    | 2.4     | <div>NOT SIGNIFICANT</div> |
| Wasatch (Alta/Snowbird/Park City)          | UT     | intermountain | 48      | 2.5    | 2.6     | +0.1 | +3%    | 2.5     | <div>NOT SIGNIFICANT</div> |
| Chugach (Alyeska)                          | AK     | maritime      | 43      | 3.1    | 3.2     | +0.1 | +4%    | 3.1     | <div>NOT SIGNIFICANT</div> |
| San Juans (Telluride/Purgatory)            | CO     | continental   | 46      | 2.2    | 2.2     | +0.1 | +4%    | 2.2     | <div>NOT SIGNIFICANT</div> |
| Spring Mtns (Lee Canyon)                   | NV     | continental   | 18      | 1.5    | 1.6     | +0.1 | +5%    | 1.3     | <div>NOT SIGNIFICANT</div> |
| Uinta Mtns                                 | UT     | continental   | 48      | 1.3    | 1.4     | +0.1 | +7%    | 1.1     | <div>NOT SIGNIFICANT</div> |
| N Sangre de Cristo (Taos)                  | NM+CO  | continental   | 47      | 0.8    | 0.9     | +0.1 | +12%   | 0.7     | <div>NOT SIGNIFICANT</div> |
| N Sierra / Tahoe (Palisades/Heavenly/Rose) | CA+NV  | maritime      | 48      | 4.3    | 4.8     | +0.5 | +12%   | 4.1     | <div>NOT SIGNIFICANT</div> |
| Park Range (Steamboat)                     | CO     | continental   | 47      | 2.1    | 2.4     | +0.3 | +12%   | 2.1     | <div>SURVIVES FDR</div>    |
| Snowy Range & Sierra Madre                 | WY     | continental   | 48      | 1.6    | 1.9     | +0.2 | +13%   | 1.6     | <div>SURVIVES FDR</div>    |
| Markagunt (Brian Head)                     | UT     | continental   | 46      | 2.5    | 2.9     | +0.3 | +13%   | 2.3     | <div>NOT SIGNIFICANT</div> |
| S Sangre de Cristo (Santa Fe)              | NM     | continental   | 46      | 1.0    | 1.2     | +0.2 | +22%   | 0.9     | <div>SURVIVES FDR</div>    |
| Tushar Mtns (Eagle Point)                  | UT     | continental   | 47      | 1.4    | 1.7     | +0.3 | +23%   | 1.3     | <div>NOT SIGNIFICANT</div> |
| White Mtns AZ (Sunrise Park)               | AZ     | continental   | 46      | 0.9    | 1.2     | +0.2 | +24%   | 0.6     | <div>SURVIVES FDR</div>    |

| SKI REGION                        | STATES | SNOW CLIMATE | WINTERS | NORMAL | EL NIÑO | DIFF | CHANGE | LA NIÑA | STATUS          |
|-----------------------------------|--------|--------------|---------|--------|---------|------|--------|---------|-----------------|
| San Francisco Peaks (AZ Snowbowl) | AZ     | continental  | 46      | 1.7    | 2.1     | +0.4 | +24%   | 1.4     | NOT SIGNIFICANT |
| Jemez (Pajarito)                  | NM     | continental  | 46      | 0.7    | 1.0     | +0.3 | +38%   | 0.7     | SURVIVES FDR    |



Powder days in a normal winter against an El Niño winter, every region with data, sorted by effect.

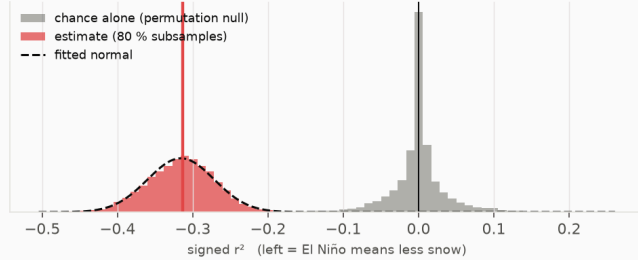
## El Niño and the ski season, by region and window of winter



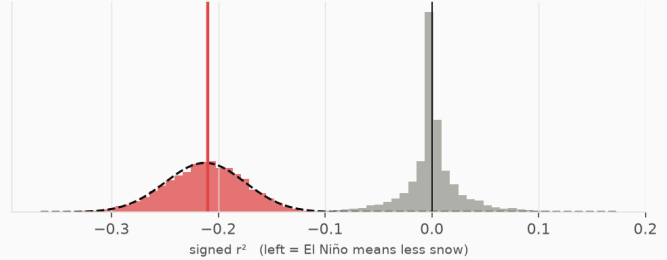
Variance explained ( $r^2$ ) by region, window and measure, regions ordered north to south. The vertical gradient is the seesaw.

## Probability distributions behind each result

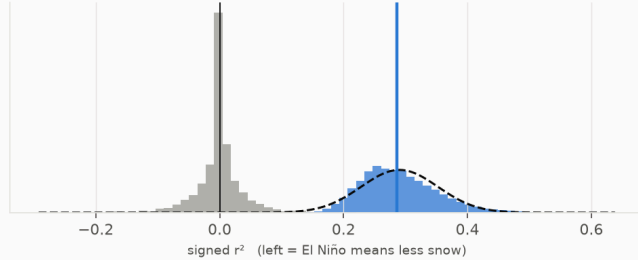
**Missoula ranges (Montana Snowbowl) — Midwinter, big storm days**  
 $r^2 = 31\%$  of year-to-year variance · El Niño = less snow ·  $n = 46$  ·  $p = 0.0002$



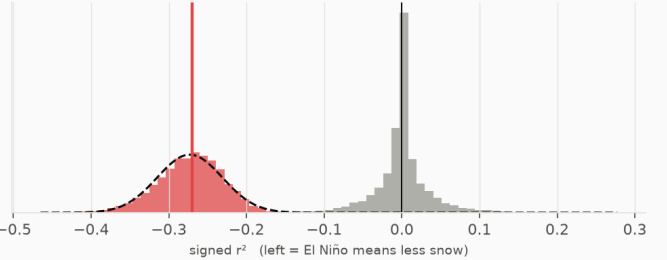
**Bitterroots (Lost Trail) — Spring, mean swe**  
 $r^2 = 21\%$  of year-to-year variance · El Niño = less snow ·  $n = 59$  ·  $p = 0.0004$



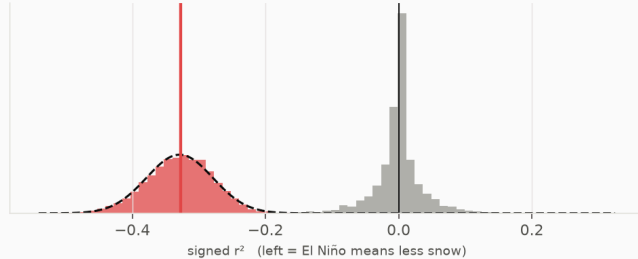
**S Sangre de Cristo (Santa Fe) — Spring, days\_with\_base**  
 $r^2 = 29\%$  of year-to-year variance · El Niño = more snow ·  $n = 46$  ·  $p = 0.0004$



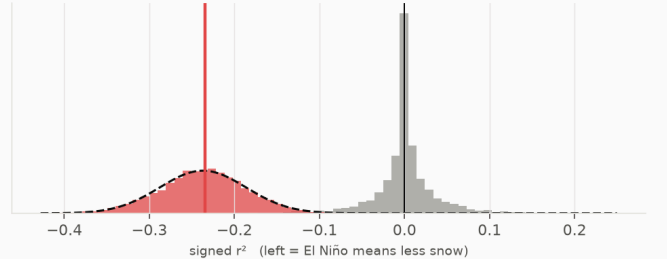
**Missoula ranges (Montana Snowbowl) — Spring, mean swe**  
 $r^2 = 27\%$  of year-to-year variance · El Niño = less snow ·  $n = 46$  ·  $p = 0.0004$



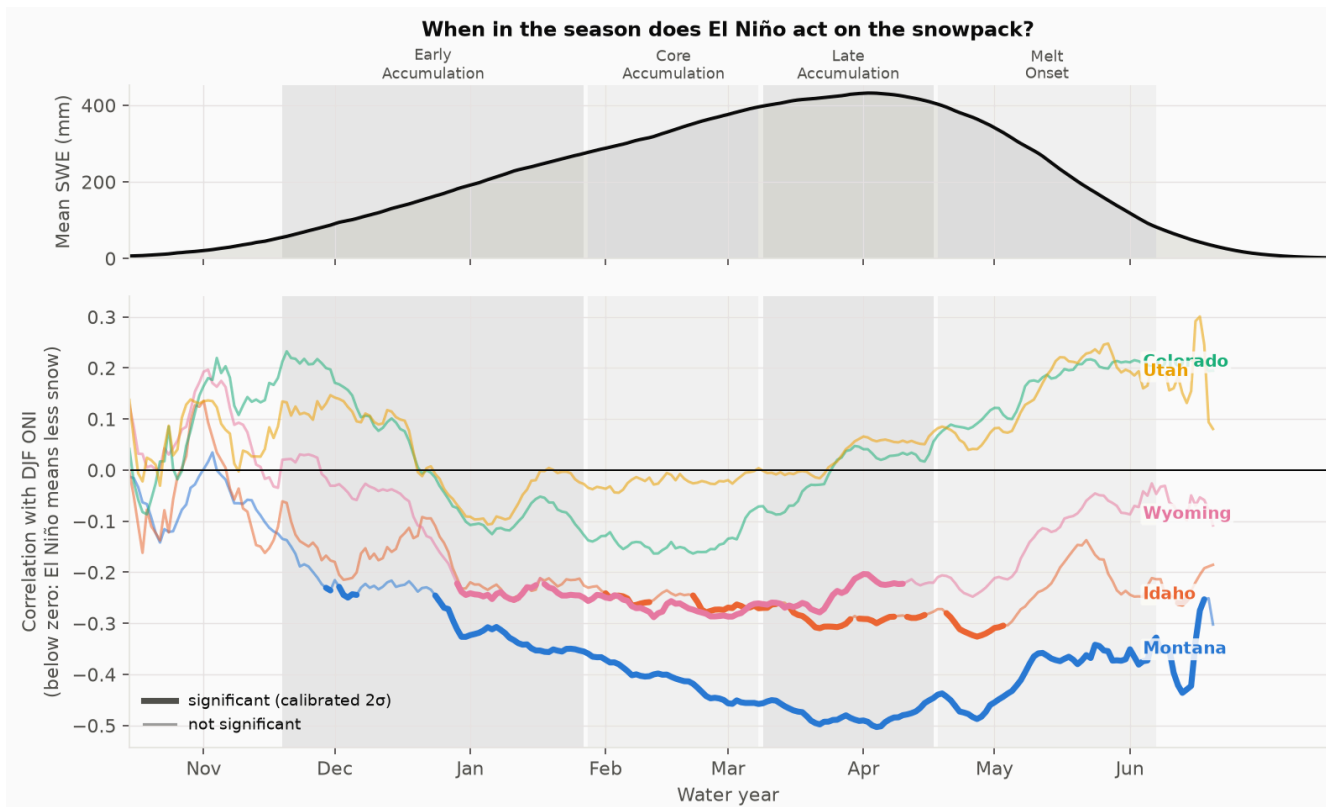
**Missoula ranges (Montana Snowbowl) — Spring, days\_with\_base**  
 $r^2 = 33\%$  of year-to-year variance · El Niño = less snow ·  $n = 46$  ·  $p = 0.0002$



**Mt Hood — Spring, mean swe**  
 $r^2 = 23\%$  of year-to-year variance · El Niño = less snow ·  $n = 48$  ·  $p = 0.0008$



Probability distributions behind selected headline results: the subsampling distribution of the estimate against the permutation null, with the fitted normal. Separation between the two is the significance, shown rather than asserted.



The correlation with the DJF index for every day of the water year, against the snowpack climatology, showing when in the season the signal acts.

## METHOD

### How this was measured

#### A powder day is six inches of snow

A fixed water threshold is a maritime yardstick. Cascade new snow runs about 12 % density and Colorado's about 7 %, so 25 mm of water is roughly eight inches of Cascade snow but fourteen inches of Colorado snow — counting water alone makes the driest, lightest snowpacks look storm-free. Each station's own new-snow ratio is measured from the overlap of its depth and water records, then applied to its much longer water record. Across 508 stations the median ratio is 7.5 inches of snow per inch of water.

#### Regions are ranges, not states

A state boundary is not a snow boundary. Tahoe spans California and Nevada, the Bitterroots and Selkirks span Montana and Idaho, the Tetons span Wyoming and Idaho, the Sangre de Cristo spans Colorado and New Mexico. Each region is anchored on a mountain range around real ski terrain and labelled with its snow climate.

#### Stations are weighted by elevation

A gauge represents a ski area only insofar as it shares its snow climate, and elevation is most of that. Each region carries its served base-to-summit band; stations are weighted by a Gaussian on the vertical gap to that band and another on horizontal distance, so a valley gauge below the lifts does not speak for a mountain.

Significance, and the correction for asking often

The modified bootstrap of Brown & Harper (2026): omit a random 20 % of winters, refit, repeat ten thousand times, and require the 2σ bounds of the resulting near-Gaussian distribution to exclude zero. That raw rule fires on about 30 % of pure noise, because the subsample spread understates the sampling error by the delete-*d* factor  $\sqrt{((n-d)/d)} = 2$ , so a calibrated version and a permutation p-value are reported alongside. Benjamini-Hochberg control is then applied across the entire grid.

Two views of the same data

The statistical tests run on per-station detrended z-scores, so a long-term warming trend cannot pose as an ENSO signal. The physical composites are raw averages by phase — what actually happened. The two are matched explicitly on region, window and measure rather than assumed to line up.

Windows that carry a season

Early season (1 Nov – 15 Dec), Holidays (16 Dec – 5 Jan), Midwinter (6 Jan – 28 Feb) and Spring (1 Mar – 15 Apr). They abut without gaps. These replace the hydrologic periods used in water-supply work, which are built around runoff timing rather than revenue.

COVERAGE

Regions and station coverage

HOW A REGION'S LOCATION IS FIXED

Each region's centre is computed from the coordinates of the ski areas it serves, never typed by hand. An earlier draft did type them, and drifted: one region's centre sat 61 km from Schweitzer, and another grouped Lookout Pass with Montana Snowbowl 138 km away under a name matching neither. The regions are now split where their areas were too far apart to share a snow climate — the Selkirks from the Coeur d'Alenes, Bridger Bowl from Big Sky, Wolf Creek from Telluride, Taos from Santa Fe — and a test checks that every named ski area falls inside a state its region claims.

Effective station count is Kish's measure applied to the elevation weights. Regions with few stations inside the served band — notably southern Utah and the Snowy Range — should be read with more caution, because SNOTEL does not reach those elevations locally.

| SKI REGION                            | STATES | CLIMATE      | SERVED BAND (FT) | STATIONS | EFFECTIVE | WEIGHTED ELEV (FT) | IN BAND |
|---------------------------------------|--------|--------------|------------------|----------|-----------|--------------------|---------|
| Beartooth & Absaroka (Red Lodge)      | MT+WY  | continental  | 7,016–9,416      | 14       | 12.4      | 8,101              | 10      |
| Bighorn Mtns (Meadowlark)             | WY     | continental  | 7,500–9,500      | 15       | 13.4      | 8,716              | 12      |
| Bitterroots (Lost Trail)              | MT+ID  | transitional | 6,400–8,000      | 13       | 10.8      | 7,000              | 5       |
| Black Hills (Terry Peak)              | SD     | continental  | 5,800–7,064      | 3        | 2.9       | 6,360              | 3       |
| Blue Mtns (Anthony Lakes)             | OR     | transitional | 7,100–8,000      | 18       | 13.7      | 5,809              | 2       |
| Bridger Range (Bridger Bowl)          | MT     | continental  | 6,100–8,700      | 6        | 5.2       | 7,156              | 6       |
| Central Cascades (Stevens/Snoqualmie) | WA     | maritime     | 3,000–5,845      | 24       | 20.9      | 3,832              | 20      |
| Central Oregon (Mt Bachelor)          | OR     | transitional | 6,300–9,065      | 10       | 7.3       | 5,220              | 0       |
| Chugach (Alyeska)                     | AK     | maritime     | 250–2,750        | 11       | 9.5       | 1,413              | 7       |

| SKI REGION                                 | STATES | CLIMATE       | SERVED BAND (FT) | STATIONS | EFFECTIVE | WEIGHTED ELEV (FT) | IN BAND |
|--------------------------------------------|--------|---------------|------------------|----------|-----------|--------------------|---------|
| Coeur d'Alenes (Silver/Lookout)            | ID+MT  | transitional  | 4,100–6,300      | 8        | 6.3       | 5,055              | 7       |
| E Cascades rain shadow (Mission Ridge)     | WA     | transitional  | 4,570–6,820      | 7        | 5.9       | 4,744              | 2       |
| Elk Mtns (Aspen/Crested Butte)             | CO     | continental   | 7,945–12,162     | 15       | 13.8      | 10,139             | 15      |
| Front Range (Winter Park/Loveland/A-Basin) | CO     | continental   | 9,000–13,050     | 26       | 22.2      | 10,403             | 22      |
| Gore & Tenmile (Vail/Summit/Copper)        | CO     | continental   | 8,120–12,998     | 25       | 22.5      | 10,388             | 25      |
| Jemez (Pajarito)                           | NM     | continental   | 8,500–10,441     | 10       | 8.2       | 9,514              | 5       |
| Madison & Gallatin (Big Sky)               | MT     | continental   | 6,800–11,166     | 10       | 9.2       | 8,173              | 10      |
| Markagunt (Brian Head)                     | UT     | continental   | 9,600–10,970     | 13       | 9.9       | 9,123              | 3       |
| Missoula ranges (Montana Snowbowl)         | MT     | transitional  | 5,000–7,600      | 7        | 6.2       | 6,013              | 5       |
| Mt Hood                                    | OR     | maritime      | 4,500–8,540      | 11       | 9.4       | 4,268              | 2       |
| N Sangre de Cristo (Taos)                  | NM+CO  | continental   | 9,200–12,481     | 13       | 11.0      | 10,169             | 12      |
| N Sierra / Tahoe (Palisades/Heavenly/Rose) | CA+NV  | maritime      | 6,200–10,067     | 27       | 24.9      | 7,573              | 26      |
| North Cascades (Mt Baker)                  | WA     | maritime      | 3,500–5,089      | 10       | 8.6       | 4,160              | 5       |
| Park Range (Steamboat)                     | CO     | continental   | 6,900–10,568     | 19       | 16.5      | 9,355              | 17      |
| Ruby Mtns & NE Nevada                      | NV     | continental   | 6,500–10,000     | 11       | 10.0      | 7,866              | 10      |
| S San Juans (Wolf Creek)                   | CO     | continental   | 10,300–11,904    | 16       | 12.4      | 10,970             | 12      |
| S Sangre de Cristo (Santa Fe)              | NM     | continental   | 10,350–12,075    | 8        | 5.9       | 10,371             | 3       |
| S Sierra (Mammoth/June)                    | CA     | maritime      | 7,953–11,053     | 2        | 1.7       | 9,036              | 2       |
| S Washington Cascades (Crystal/White Pass) | WA     | maritime      | 4,400–7,012      | 24       | 18.8      | 4,683              | 12      |
| Sacramento Mtns (Ski Apache)               | NM     | continental   | 9,600–11,500     | 1        | 1.0       | 10,250             | 1       |
| San Francisco Peaks (AZ Snowbowl)          | AZ     | continental   | 9,200–11,500     | 8        | 4.8       | 8,422              | 1       |
| San Juans (Telluride/Purgatory)            | CO     | continental   | 8,725–13,150     | 23       | 21.6      | 10,381             | 23      |
| Sawtooth & Smoky (Sun Valley)              | ID     | intermountain | 5,750–9,150      | 21       | 18.6      | 7,514              | 18      |
| Selkirks (Schweitzer)                      | ID     | transitional  | 4,000–6,400      | 9        | 7.2       | 5,319              | 8       |
| Snowy Range & Sierra Madre                 | WY     | continental   | 8,798–9,663      | 22       | 18.7      | 9,475              | 7       |
| Spring Mtns (Lee Canyon)                   | NV     | continental   | 8,510–11,290     | 4        | 4.0       | 8,723              | 3       |
| Tetons (Jackson/Targhee)                   | WY+ID  | intermountain | 6,300–10,450     | 11       | 9.3       | 7,874              | 11      |
| Tushar Mtns (Eagle Point)                  | UT     | continental   | 9,000–10,600     | 11       | 9.4       | 9,359              | 7       |
| Uinta Mtns                                 | UT     | continental   | 8,000–11,000     | 30       | 27.9      | 9,345              | 26      |
| W Central Idaho (Brundage/Tamarack)        | ID     | intermountain | 5,840–7,640      | 11       | 9.9       | 5,937              | 6       |
| Wasatch (Alta/Snowbird/Park City)          | UT     | intermountain | 6,800–11,000     | 30       | 25.2      | 8,173              | 28      |
| White Mtns AZ (Sunrise Park)               | AZ     | continental   | 9,200–11,100     | 9        | 7.9       | 8,673              | 2       |
| Whitefish & Flathead Range                 | MT     | transitional  | 4,464–6,817      | 12       | 10.9      | 5,443              | 10      |
| Wind River Range                           | WY     | continental   | 7,000–10,000     | 27       | 23.7      | 8,956              | 24      |

### LIMITS

## What this does not establish

- An  $r^2$  of 15–35 % is real and it is not a forecast. ENSO shifts the odds for a season. Most of the year-to-year variance remains weather.

- **Snow depth records begin around 1993.** New-snow ratios are calibrated on that period and applied backward, which assumes density has been stable. Depth-based composites rest on roughly 25 winters against 45 to 60 for the water-based ones.
- **A SNOTEL gauge is not a ski area.** Even with elevation weighting these are nearby mountain stations, and they say nothing about aspect, grooming or snowmaking.
- **Overlapping regions are not fully independent** where their radii intersect, so the sign tests read slightly optimistically.
- **Six inches is a choice.** Regions with few powder days in absolute terms show large percentage swings on small differences.
- **The strength null is bounded by power.** Within-phase tests had about sixteen winters and could only have detected a correlation above roughly 0.65. The claim is that across 2508 chances nothing rose above noise, and the better-powered nested test also found nothing.

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Sources – NOAA Climate Prediction Center, Oceanic Niño Index. USDA NRCS Air & Water Database, SNOTEL daily snow water equivalent, snow depth, precipitation and temperature. NOAA NCEI nClimDiv. Method after J. Brown and J. Harper (2026), *Historical evolution of snowpack capacity to buffer rain-on-snow runoff in a large Columbia River headwaters basin*, Hydrol. Earth Syst. Sci. 30, 5735–5748. False-discovery control after Benjamini & Hochberg; field significance after Wilks (2016).

Analysis code and every table in this report are reproducible from [github.com/yetiskier/enso-snowpack](https://github.com/yetiskier/enso-snowpack).